Universidade do Estado do Rio de Janeiro
Centro de Ciências Sociais
Instituto de Estudos Sociais e Políticos

First name Last name

Title of the dissertation or thesis

Rio de Janeiro
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First name Last name

Title of the dissertation or thesis

Dissertation presented, as a partial requirement for obtaining the degree of Master, to the Graduate Program in Political Science, Universidade do Estado do Rio de Janeiro.

Advisor: Prof. Dr. First name Last name

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Dedication text.

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ABSTRACT

LAST NAME, F. *Title of the dissertation or thesis*. yyyy. 14 f. Dissertação (Mestrado em Ciência Política) – Instituto de Estudos Sociais e Políticos, Universidade do Estado do Rio de Janeiro, Rio de Janeiro, yyyy.

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RESUMO

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Texto do resumo em português. O resumo deve conter entre 150 e 500 palavras conforme as normas da UERJ.

Palavras-chave: Primeira palavra-chave. Segunda palavra-chave. Terceira palavra-chave.

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INTRODUCTION

> To the worm that first gnawed the cold flesh of my corpse I dedicate these Posthumous Memoirs as a fond remembrance (Assis, 1881).

In statistics, “all models are wrong, but some are useful” (Box, 1976). A model is a deliberate simplification of reality, and the craft of modelling lies in choosing which aspects to keep. Gelman; Hill (2007) show how regression and hierarchical models negotiate this tension between parsimony and realism.

The quality of our conclusions depends on both the model and the data. A striking example is the forecasting of U.S. elections from a sample of Xbox users — heavily biased relative to the electorate — which, after suitable statistical adjustment, produced estimates competitive with traditional polls (Wang *et al.*, 2015).

1 ORDINARY LEAST SQUARES

Consider the linear regression model in matrix form,

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon, \tag{1}$$

where $\mathbf{y}$ is the $n \times 1$ response vector, $\mathbf{X}$ is the $n \times p$ design matrix, β is the coefficient vector and ε is the error vector.

1.1 Closed-form derivation

The ordinary least squares estimator minimises the residual sum of squares:

$$S(\beta) = \|\mathbf{y} - \mathbf{X}\beta\|^2 = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta).$$

Expanding the product and using that $\beta^\top \mathbf{X}^\top \mathbf{y}$ is a scalar (hence equal to its transpose),

$$S(\beta) = \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta.$$

The first-order condition follows by differentiating with respect to β and setting the gradient to zero:

$$\frac{\partial S}{\partial \beta} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\beta \stackrel{!}{=} \mathbf{0}.$$

Assuming $\mathbf{X}^\top \mathbf{X}$ is invertible, we solve for the estimator in its celebrated closed form:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \tag{2}$$

1.2 Illustration

Figure 1 shows a set of points (x_i, y_i) and the line fitted by Equation 2. Table 1 summarises the estimates, with their standard errors and t-statistics.

Table 1 - Ordinary least squares estimates. Source: prepared by the author (2026).

Coefficient	Estimate	Std. error	t-statistic
β_0 (intercept)	1.78	0.25	7.20
β_1 (slope)	0.79	0.04	18.95

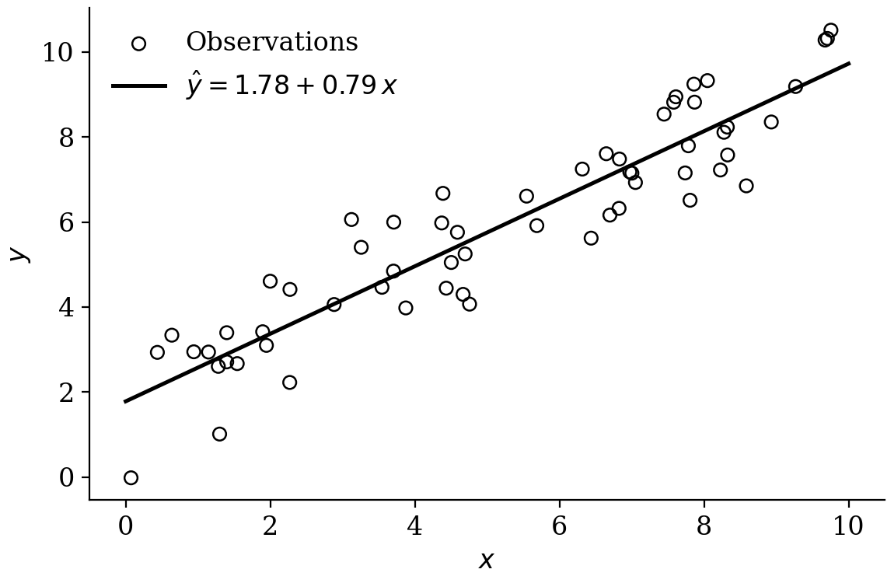

Figure 1 - Regression line fitted by ordinary least squares. Source: prepared by the author (2026).

CONCLUSION

Models are useful when they help us learn from data (Box, 1976), and the closed form of OLS in Equation 2 is the starting point for richer extensions, such as the hierarchical models of Gelman; Hill (2007).

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